

Example frozen decision card

Frozen card: CARA_LO_CHEMBL2328568_IC50_0001 | source task: CHEMBL2328568_IC50

card header

task_id: CARA_LO_CHEMBL2328568_IC50_0001
assay_context: {source: CARA, assay_id: CHEMBL2328568_IC50}

card-level metadata

budget_k: 10; support_size: 50; candidate_pool_size: 967; feasible_candidate_count: 618

support_set excerpt

Support activity and descriptors are stored in the card and are visible to deployable systems.

id	smiles	activity_value	descriptors
CHEMBL109...	CC(C)(C)c1[nH]c2cccc2c1[C@@H]1...	6.44	mw=449.6; cLogP=5.5; TPSA=68.4; HBD/HBA=3/3; rot=5
CHEMBL120591	Cc1c(F)ccc2[nH]c(C(=O)N3CCN(C)C...	8.15	mw=276.3; cLogP=1.4; TPSA=52.2; HBD/HBA=1/3; rot=1

candidate_pool excerpt

Candidate descriptors are stored in the card. Candidate activity is retained only for offline scoring.

id	smiles	activity_value	descriptors
CHEMBL113816	COc1cc(CN(C)c2ccc3nc(N)nc(N)c3c...	scorer-only	mw=339.4; cLogP=2.4; TPSA=99.5; HBD/HBA=2/7; rot=5
CHEMBL113999	COc1cc(CNc2ccc3nc(N)nc(N)c3c2)c...	scorer-only	mw=355.4; cLogP=2.4; TPSA=117.5; HBD/HBA=3/8; rot=6

hard_constraints

id	type	check	params
exact_budget	output	exactly_k	
candidate_pool_only	output	in_candidate_pool	
no_duplicates	output	no_duplicates	
exclude_support	output	exclude_support	
mw_max_500	candidate	descriptor_max	descriptor=mw; max=500
clogp_max_4_5	candidate	descriptor_max	descriptor=clogp; max=4.5
forbidden_substructures	candidate	forbidden_smarts	smarts=[]

output_schema

field	expected value
candidate_id	string
confidence	optional number
rank	integer

leakage boundary

Support activity is visible input. Candidate activity is stored in the frozen card but hidden from evaluated systems and used only by the scorer.